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A comparative study on two explosive acetone peroxides

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ABSTRACT

Two explosive cyclic acetone peroxides, diacetone diperoxide (DADP) and triacetone triperoxide (TATP) have been studied in respect of thermal decomposition, burning behavior, impact sensitivity, and initiating efficiency. Using the glass Bourdon gauge technique, the vapor pressures of TATP and DADP were determined over the temperature range 75–144 °C and 67–120 °C, respectively. The kinetic parameters of decomposition of the peroxides in the gas phase have been obtained in the temperature interval of 140–200 °C. The decomposition of both DADP and TATP followed the first-order reaction to high degrees of decay with close activation energies of 159.2 kJ/mol (38.0 kcal/mol) and 165.8 kJ/mol (39.6 kcal/mol), respectively. The decomposition rate constants of DADP were found to be approximately 2 times less than those of TATP. The linear burning rate of DADP measured in a constant-pressure window bomb appeared to be approximately 5 times less than that of TATP. Temperature profiles in the combustion wave were measured at subatmospheric pressures with the help of thin tungsten-rhenium thermocouples. The leading reaction on combustion of both volatile peroxides was assumed to occur in the gas phase. Kinetic parameters of the leading reaction derived from the combustion data showed a good agreement with kinetic parameters of low-temperature thermal decomposition extrapolated to the high-temperature flame zone. In the drop-weight impact test, DADP appeared to be notably less sensitive peroxide than TATP. No deflagration-to-detonation transition was observed when RDX was attempted to explode by using 0.5 g of DADP in the standard detonator No. 8 shell. Comparing explosive properties of the studied peroxides, DADP may be considered as a less dangerous, however still unsafe explosive in handling as TATP.

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1. Introduction

Many of organic peroxide compounds find application in the polymer industry; the other major usage is bleaching and antibacterial action. Depending on the peroxide unit concentration in the molecule, the organic peroxides may possess explosive properties to a greater or lesser extent. Explosive peroxides did not find civilian or military applications for various reasons, mainly due to high sensitivity, volatility, and instability. Unfortunately, the ease of synthesis of these materials from readily available chemicals made them popular explosives among criminals and extremists worldwide. The cyclic acetone peroxides are characteristic examples.

As a rule, the reaction of acetone with hydrogen peroxide leads to formation of two cyclic structures, diacetone diperoxide (DADP) and triacetone triperoxide (TATP) [1–4].

Monomer of acetone peroxide, dimethyldioxirane (DMDO), is often used as an oxidizer in reactions with amines and alkenes in organic chemistry [5–7]. Because of its low thermal stability DMDO is utilized in the form of dilute solutions. According to Ref. [8], the half-life time of DMDO at room temperature is 26 days.

Chinese researchers [9] published a synthesis route to tetrameric acetone peroxide (TetrATP). The synthesis was carried out in the presence of tin (IV) chloride to give 44.7% yield of TetrATP. Tetrameric acetone peroxide was identified by molecular weight determination with the help of osmometry. However, the spectra obtained with other methods such as mass- and IR-spectroscopy, ^{1}H and ^{13}C NMR spectrometry gave spectra that proved to be very close to analogous spectra of TATP. Moreover, Matyáš and Pachman in their study of effect of reaction conditions on the composition of forming product [10] have shown that using tin (IV) chloride as the reaction catalysis leads to principally the same composition as with ordinary acidic catalysis. The structural formulas of the cyclic acetone peroxides are shown in Fig. 1.

AM1 semi-empirical calculation affords a rough estimation of the O—O bond lengths in all of acetone peroxides (Table 1). The longest bond is in the strained three-membered ring DMDO; the shortest – in the six-membered dimer DADP. Although the semi-empirical calculation results are different from X-ray analysis data [11], the tendency in the variation of bond lengths is obvious. Using the experimental and calculated data for DADP and TATP, the calculated bond lengths for DMDO and TetrATP can be adjusted for this

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Table 1
Comparison of O—O bond lengths in cyclic acetone peroxides and decomposition rate constants at room temperature.

Peroxide	O—O bond length (AM1 calc.) (Å)	O—O bond length (exp.) (Å)	O—O bond length (adjusted calc.) (Å)	k_{20} (s^{-1})
DMDO	1.343		1.520	6.30E−07 [8]
DADP	1.290	1.458 [11]	1.458	9.75E−16 [12]
TATP	1.300	1.473 [11]	1.473	2.96E−14[13]
TetrATP	1.312		1.485	3.5E−12 (estm.)

difference (see Table 1). It is evident that the thermal stability of the peroxides is determined by the strength of O—O bond. One may expect, therefore, a correlation between the peroxide bond length and the decomposition rate constant. Rate constants of DADP and TATP decomposition at room temperature can be calculated from kinetic data published in [12,13], and a rate constant of DMDO decomposition can be estimated from its half-life period [8].

As seen from Fig. 2, there is a good correlation between the decomposition rate constants of the acetone peroxides at room temperature and the O—O bond length. It follows from the correlation that the tetramer, if really obtained, is two orders of magnitude less stable than the trimer.

DADP and TATP are two stable cyclic acetone peroxides available presently. In the last few years, many researches on properties of the acetone peroxides have evolved. However, DADP is still not completely studied and its properties published are frequently inconsistent. In this connection, the purpose of our work was a comparative investigation of these two peroxides. Preliminarily results of this study were presented in 2009 [14].

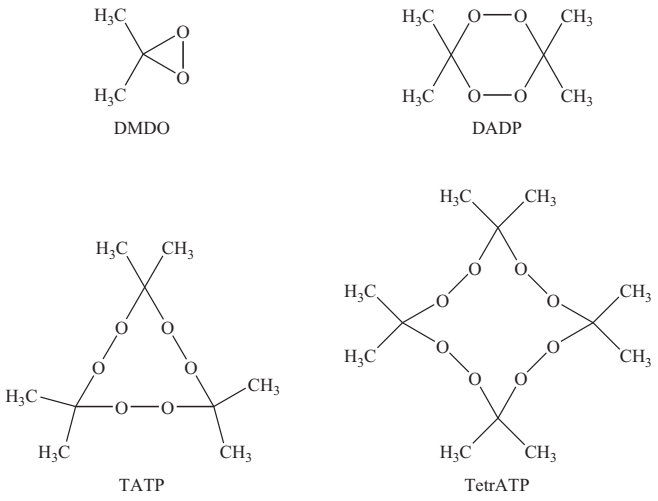


Fig. 1. Structural formulas of cyclic acetone peroxides.

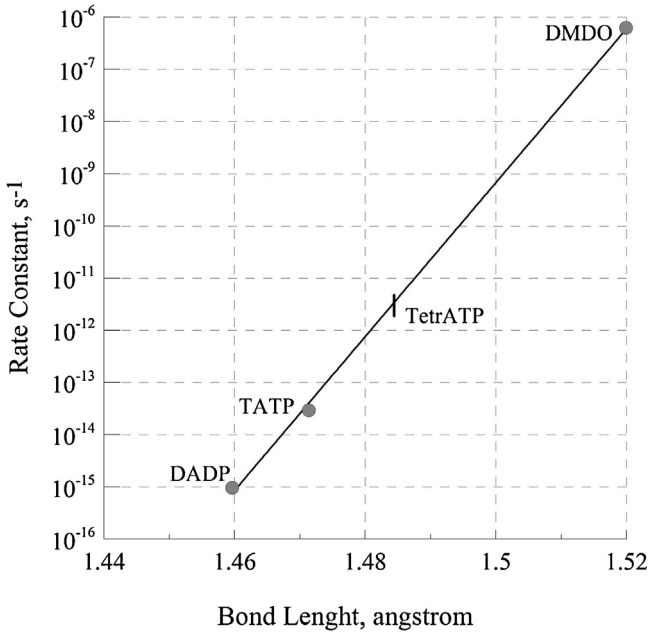


Fig. 2. Correlation between decomposition rate constants at 20 °C and O—O bond lengths for cyclic acetone peroxides.

2. Experimental

2.1. Preparation

Triacetone triperoxide was synthesized by the standard method published elsewhere [1,15] from acetone and 30% hydrogen peroxide with addition of hydrochloric acid as the reaction catalyst. Dissolving TATP in chloroform with a catalytic amount of p-toluenesulfonic acid and keeping the solution at room temperature for a week afforded pure crystalline DADP [11]. Both substances were recrystallized from hot ethanol. The purity of DADP and TATP was checked with high-performance gas–liquid chromatography combined with selective mass spectrometer. Melting points were run in sealed capillary tubes at 3 °C/min on a Boatius heating microtable and were 95–97 °C for TATP (lit. 94–98.5 °C [1,2] and 132–133 °C for DADP (lit. 131–133 °C [1,16]).

2.2. Decomposition study

Manometric experiments were carried out in thin-walled glass manometers of compensation type (the glass Bourdon gauge) at 140–202 °C. A sample of peroxide (10–15 mg) was loaded into a vial of 12–15 cm^3 volume. The vial was vacuumized up to 0.1 mm Hg on cooling by liquid nitrogen, sealed, and immersed into a thermostat with the Wood's metal. Temperature in the thermostat was maintained constant with an accuracy of ±0.5 °C. Pressure of gases evolved during the course of experiments (the accuracy of pressure measurements was ±0.5 mm Hg) was recalculated to the volume of gases at normal conditions, and the rate of gas evolution was then evaluated. The rate constant was calculated as a ratio of the initial gas evolution rate to the final volume of evolved gases.

2.3. Vapor pressure measurements

The glass Bourdon gauges were used to measure TATP and DADP vapor pressures at different temperatures. A sample of peroxide (20–50 mg) was loaded into a vial of 12–15 cm^3 volume. The vapor pressure over a peroxide sample at a given temperature was fixed as the gauge reading after 5 min of warming-up the vial in the thermostat assuming that no perceptible decomposition had time to take place. Extracting the vial from the thermostat and cooling it to a room temperature allowed assessment of whether the decomposition took place. Usually the maximum remaining pressure in the vial after cooling did not exceed 2 mm Hg, while the full decomposition of the sample resulted in 200–300 mm Hg pressure of cooled gases. Results of vapor pressure measurements are presented in Table 2.

Table 2
TATP and DADP vapor pressure data.

TATP		DADP	
Temperature (°C)	Vapor pressure (mm Hg)	Temperature (°C)	Vapor pressure (mm Hg)
75.0	3.5	67.5	8.5
80.0	5.0	70.0	10.5
80.0	6.0	70.0	12.5
85.0	8.5	70.5	12.8
90.0	11.5	70.5	9.0
90.0	16.0	73.0	12.5
94.0	19.5	76.5	17.5
98.5	21.0	77.5	15.0
98.5	24.5	78.0	18.8
99.0	27.5	80.0	25.0
100.5	29.5	82.0	25.0
105.5	36.0	84.5	31.6
106.5	36.5	85.0	30.0
110.5	44.0	85.5	33.5
110.6	45.5	88.0	41.2
111.0	42.0	90.0	48.5
114.8	51.0	94.5	59.1
117.0	50.5	98.5	82.0
120.0	62.0	105.0	126.5
120.0	65.0	110.3	168.0
121.0	66.5	114.3	204.5
123.9	76.5	119.8	260.0
125.0	85.5		
126.0	81.0		
128.5	94.0		
132.7	115.5		
137.2	143.0		
142.7	177.0		

2.4. Combustion study

Burning rates of DADP were measured in a window constant-pressure bomb of 1.5-liter volume in the 0.1–2 MPa pressure interval using a high-speed video camera. Samples to test were prepared as pressed cylinders of 0.95–0.97 TMD confined in transparent acrylic tubes of 4 mm internal diameter.

Temperature profiles in the combustion wave were measured using Π-shaped micro thermocouples. The thermocouples were welded from 80%W + 20%Re and 95%W + 5%Re wires 25 μm thick, rolled in bands 5–7 μm thick, and configured to a Π-shape. The thermocouple was embedded into the center of the sample so that the section with the junction was parallel to the combustion front. The thermocouple signal was recorded with a Pico ADC 216 digital oscilloscope.

2.5. Impact sensitivity measurements

Impact sensitivity of DADP and TATP was measured on Russian K-44-1 impact machine for primary explosives. A weight of 200 g was dropped onto a 3 mm steel striker set over 12 mg pressed sample. Bruceton analysis method was used to determine H_{50} value.

2.6. Initiating efficiency tests

The ability of DADP and TATP to initiate high explosive, RDX, was tested in detonator No. 8 brass caps of 7 mm e.d. and 0.3 mm wall thickness. RDX (0.7 g) was first pressed into the shell at 100 MPa followed by pressing a weight of peroxide and a reinforcing cap at 30 MPa. The reinforcing cap was made from brass and had 1.2 mm ignition orifice. A detonator was mounted on a 4 mm thick lead plate serving as a witness. The ignition was performed with an electric igniter.

Thermodynamic calculations were performed with the use of the "REAL" computer simulation of chemical equilibrium code [17].

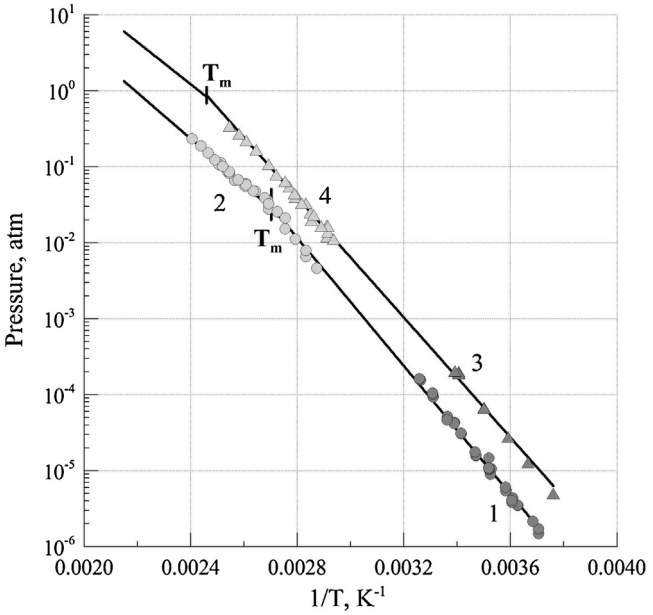


Fig. 3. Temperature dependence of vapor pressure for TATP (1 – [24], 2 – this work) and DADP (3 – [24], 4 – this work). T_m is melting point of peroxides. Solid lines are fittings of our data and data [24].

3. Results and discussion

3.1. Vapor pressure and thermodynamic parameters for TATP and DADP

The temperature dependence of TATP vapor pressure has been obtained by Oxley et al. [18] in the interval of 12–58 °C using the gas chromatography headspace technique in 2005. Based on the dependence, the heat of sublimation was found to be 104.5 kJ/mol (25.0 kcal/mol). Despite higher volatility of TATP in comparison with trinitrotoluene, the heats of sublimation of both substances were surprisingly noted to be nearly the same.

In 2009, we measured TATP vapor pressure directly above both solid and liquid substance in the overall temperature interval 75–144 °C using Bourdon gauge (Fig. 3). The results obtained were in a good agreement with low-temperature Oxley's data and allowed making some correction for the heat of TATP sublimation from one hand, and defining the heat of evaporation on the other hand. In the temperature interval of 12–97 °C, the temperature dependence of vapor pressure over solid TATP can be described as $\ln P(\text{atm}) = 22.73 - 9695.5/T$, which gives an adjusted value of sublimation heat of 80.6 kJ/mol (19.2 kcal/mol). After melting, the temperature dependence of vapor pressure is described as $\ln P(\text{atm}) = 15.29 - 6978.8/T$, with evaporation heat of 58.0 kJ/mol (13.9 kcal/mol). As seen from Fig. 3, there is an inflection point on the temperature dependence of vapor pressure at the melting point. The difference between slop angles of the fitting lines before and after melting point affords the heat of melting, 22.6 kJ/mol (5.4 kcal/mol). This value is close to data of Oxley et al. [13] obtained from DSC analysis, 24.2 kJ/mol (5.8 kcal/mol).

In 2009, Oxley et al. [19] published results of a new study, in which the heat of TATP sublimation was measured as 73.0 kJ/mol (17.4 kcal/mol). In work of Félix-Rivera et al. [20], gravimetric analysis was used to obtain TATP vapor pressure in the interval of 30–65 °C, giving the heat sublimation as 72.1 kJ/mol (17.2 kcal/mol). By using the same technique, Brady et al. reported later the heat of TATP sublimation as 83.8 kJ/mol (20.0 kcal/mol) [21].

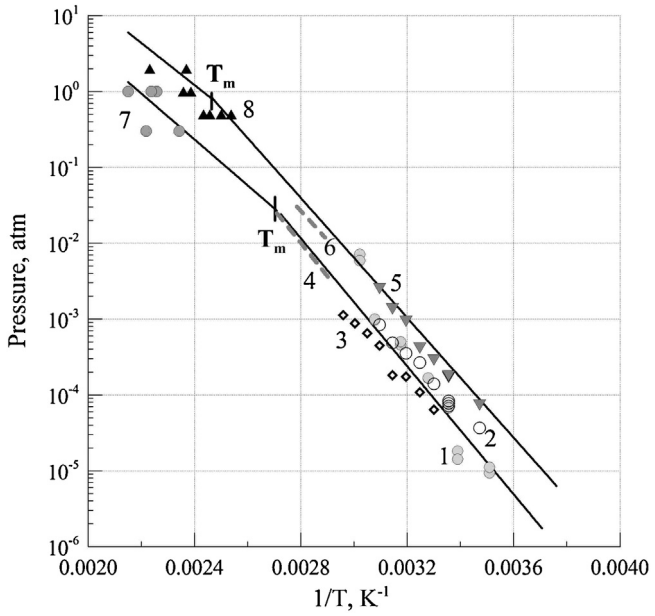


Fig. 4. Temperature dependence of vapor pressure for TATP (1) [18], (2) [19], (3) [20], dashed line (4) [21] and DADP (5) [19], dashed line (6) [21]. Solid lines are fittings of our data and data [24]. Shown also are burning surface temperatures of TATP (7) and DADP (8) measured in thermocouple-aided experiments.

TATP was also studied by means of spectroscopic techniques that provided only relative vapor pressures as a function of temperature. Cavity ringdown spectroscopy measurements, reported in work by Steinfeld et al. [22], yielded the heat of TATP sublimation in the temperature interval of −5 to 30 °C as 75.1 kJ/mol (17.9 kcal/mol). Dunayevskiy et al. [23] used photoacoustic spectroscopy with a tunable infrared quantum cascade laser to make measurements over the temperature range of −30 to 30 °C. Dunayevskiy et al. reported the heat of TATP sublimation as 86 kJ/mol (20.6 kcal/mol), but due to some systematic error for low-temperature measurements, discovered in work by Damour et al. [24], a new fit to their data points without five lowest ones yielded a value of 81.3 kJ/mol (19.4 kcal/mol).

The vapor pressure of TATP over the temperature range of −3.3 to 36.8 °C was measured by Damour et al. [24] using a modified Knudsen effusion apparatus. The Clausius–Clapeyron expression yielded the heat of sublimation of 86.2 kJ/mol (20.6 kcal/mol).

As presented in Table 3, the heats of TATP sublimation measured by different techniques vary from 72.1 to 104.5 kJ/mol. If one describes data points on vapor pressure above solid TATP, obtained by direct measurements in our work and work of Damour et al. [24], the resulting dependence over a wide temperature range of −3.3 to 97 °C will be as $\ln P(\text{atm}) = 22.73 - 9709.1/T$ with the heat of sublimation of 80.8 kJ/mol (19.3 kcal/mol) (Fig. 3). This dependence is in a good agreement with all the other data sets (Fig. 4).

Using Bourdon gauge technique, DADP vapor pressure over the solid substance was first measured in the interval of 67–120 °C in our group in 2009. DADP proved to be more volatile substance than TATP (see Fig. 3). The temperature dependence of vapor pressure is described as $\ln P(\text{atm}) = 20.82 - 8569.1/T$, with the heat of sublimation of 71.3 kJ/mol (17.0 kcal/mol). No measurements were conducted at temperatures above DADP melting point because of possible decomposition of the peroxide.

Oxley et al. also published temperature dependence of DADP vapor pressure in the interval of 12–55 °C in 2009 [19] measured by using gas chromatography headspace technique. The heat of sublimation was found to be 81.0 kJ/mol (19.4 kcal/mol). In work [21], gravimetric analysis was used for measuring DADP vapor pressures of in the interval of 57–96 °C and the heat of DADP sublimation was obtained as 72.9 kJ/mol (17.4 kcal/mol).

The modified Knudsen method employed in work [24] yielded the heat of DADP sublimation of 84.6 kJ/mol (20.2 kcal/mol) over the temperature range of −7.3 to 21.7 °C.

Data on DADP sublimation heats obtained by different techniques are collected in Table 3 and vary from 71.3 kJ/mol to 84.6 kJ/mol. The direct measurements of DADP vapor pressures performed in our work and work of Damour et al. [24] may be combined together to give a general dependence as $\ln P(\text{atm}) = 22.22 - 9091.4/T$ for a wide temperature interval (−7.3 to 120 °C) (Fig. 3), from which the heat of sublimation is calculated as 75.6 kJ/mol (18.6 kcal/mol). The data points of other authors are closely grouped along this dependence (Fig. 4).

Assuming the heat of DADP melting as 24.7 kJ/mol (5.9 kcal/mol), the heat of evaporation can be adopted as 53.1 kJ/mol (12.7 kcal/mol), and the temperature dependence of vapor pressure over liquid DADP will be as $\ln P(\text{atm}) = 15.54 - 6392/T$.

The above temperature dependences afford TATP and DADP boiling points at atmospheric pressure as 183 and 138 °C, respectively.

3.2. Thermal decomposition of TATP and DADP

Thermal decomposition of cyclic acetone peroxides has received much attention in the literature. In work [13], decomposition of TATP was investigated in the temperature interval of 151–230 °C. The decomposition kinetics was found to follow the first-order reaction to high degrees of conversion. The rate constant was described as $k\,(\text{s}^{-1}) = 3.75 \cdot 10^{13} \cdot \exp(-18{,}160/T)$.

Decomposition of a TATP toluene solution of initial concentration of 0.021 mol/L was studied in work [25] in the interval of 130–160 °C. The decomposition was found to follow the first-order reaction up to 78% of TATP conversion, with the rate constant described as $\ln k\,(\text{s}^{-1}) = (39.1 \pm 3.9) - (21{,}470 \pm 805)/T$.

DADP thermal decomposition was studied both in the gas phase in the presence of n-octane [12] and in solutions [26,27]. Decomposition kinetics in the gas phase at 170–215 °C and in the benzene solutions at 135.5–165.0 °C followed the first-order reaction to large degrees of conversion (60–70%) and were described as $\ln k\,(\text{s}^{-1}) = (31.8 \pm 2.5) - (19{,}630 \pm 1260)/T$ and $\ln k\,(\text{s}^{-1}) = (30.8 \pm 2.6) - (18{,}420 \pm 263)/T$, respectively.

In the present work, experiments on decomposition of TATP and DADP were carried out in thin-walled glass Bourdon gauges at the load density (a ratio of sample mass to the gauge volume, m/V) of 0.6–2.1 mg/cm^3. At the experimental temperatures and loads, the samples of TATP and DADP were totally sublimated in the gas phase.

The curves of the gas release in decomposition of TATP and DADP are presented in Figs. 5 and 6. The full decomposition produces 565 cm^3/g or about 5.7 moles of gases per one mole of TATP and 535 cm^3/g or 3.5 moles of gases per one mole of DADP. Cooling the remaining gases to room temperatures resulted in condensation of 12% of gases in the case of TATP and 25% of gases in the case of DADP.

The rate of the decomposition was found to depend on the initial concentration of peroxide, showing a slight increase with growing concentration. Thus, a double increase in the initial concentration (vapor pressure) resulted in 40% increase in the rate constant (Table 4). In this connection, only experimental data obtained at close initial concentrations were used in the calculations of the decomposition kinetic parameters.

Rate constants have been derived from the gas pressure change, which can be described by an equation of the first reaction order up to at least 80% of decomposition degree

Table 3
Heats of sublimation and vapor pressures for TATP and DADP.

References	TATP		DADP	
	ΔH_{sub} (kJ/mol)	Vapor pressure at 25 °C (Pa)	ΔH_{sub} (kJ/mol)	Vapor pressure at 25 °C (Pa)
Steinfeld et al. [22]	75.1 ± 4.7			
Oxley et al. [18]	104.5 ± 13.0	5.7		
Dunayevskiy et al. [23]	81.3 ± 2.6			
Our data [14]	80.6	5.6	71.3	36.3
Oxley et al. [19]	73.0	7.85	81.0	17.89
Félix-Rivera et al. [20]	72.1			
Damour et al. [24]	86.2 ± 1.5	6.0	84.6 ± 2.6	17
Oxley et al. [21]	83.8 ± 0.2	3.7	72.9 ± 2.9	19.27
Combined data [14,24]	80.7	5.3	76.5	25.5

Table 4
The kinetic parameters of TATP and DADP thermal decomposition in the gas phase.

Temperature (°C)	TATP		DADP	
	P_0 (mm Hg)	$k \cdot 10^5$ (s^{-1})	P_0 (mm Hg)	$k \cdot 10^5$ (s^{-1})
140	92	0.17	142	0.081
150	109.5	0.44		
151	156[a]	0.56	116	0.25
160	157[a]	1.53		
170	86.5	3.95		
180	84	9.75	130	3.97
180	198.5[a]	14.21		
190	109	29.91		
194.5			224.5	16.96
194.5			424.5[a]	22.72
201.5			167.5	32.36
Kinetic parameters	$k = 1.41 \cdot 10^{15} \cdot \exp(-19{,}940/T)$		$k = 1.03 \cdot 10^{14} \cdot \exp(-19{,}150/T)$	
Coef. of determination	0.9984		0.9993	

[a] Data not taken into account in calculation of the kinetic parameters.

$P = P_\infty(1 - \exp^{-kt}) + P_0 \cdot \exp^{-kt}$, where P_0 and P_∞ are initial vapor pressure and final pressure of decomposition products. In $\ln k$ vs. $1/T$ coordinates, the experimental points for the peroxides fall on straight lines described by the following equations: k (s^{-1}) = $1.41 \cdot 10^{15} \cdot \exp(-19{,}940/T)$ for TATP and k (s^{-1}) = $1.03 \cdot 10^{14} \cdot \exp(-19{,}250/T)$ for DADP. Experimental results in comparison with data of Oxley et al. [13] for TATP and data of Cafferata et al. [12] for DADP are shown in Fig. 7.

Generally, our results for TATP are close to Oxley's data, but because of some discrepancy at low temperatures, the activation energy appears to be somewhat higher, 165.8 kJ/mol (39.6 kcal/mol) as compared to 151.9 kJ/mol (36.3 kcal/mol)

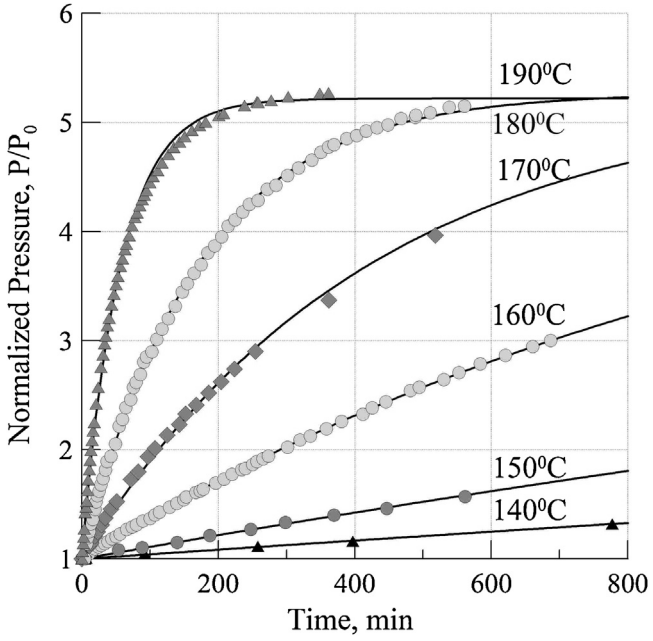


Fig. 5. Normalized pressure changes in the glass Bourdon gauge on decomposition of TATP at different temperatures. The solid lines are first-order reaction descriptions of the experimental points.

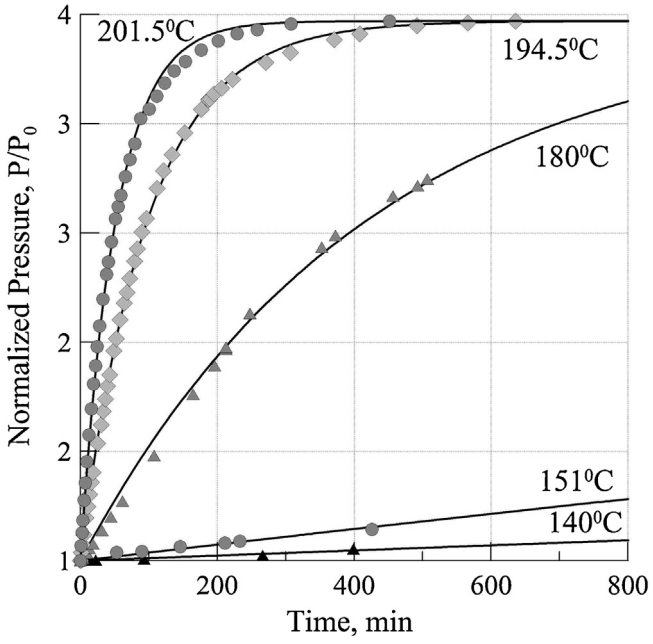


Fig. 6. Normalized pressure changes in the glass Bourdon gauge on decomposition of DADP at different temperatures. The solid lines are first-order reaction descriptions of the experimental points.

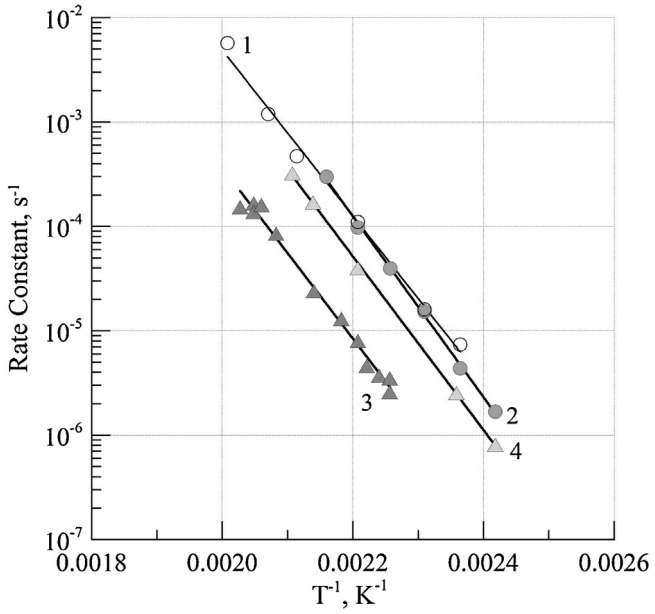


Fig. 7. Comparison of decomposition rate constants for TATP (1 – [13], 2 – this work) and DADP (3 – [12], 4 – this work).

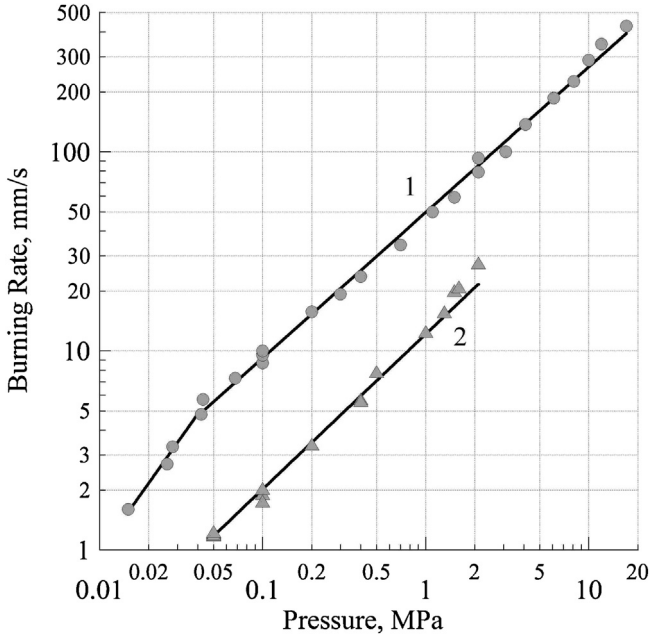


Fig. 8. Comparison of burning rates of TATP (1) and DADP (2).

obtained in [13]. Comparison of our experimental results for DADP with data of Cafferata et al. [12] shows that our rate constants are approximately 6 times higher than Cafferata's data. Probably the reason for this discrepancy may be in using neat substance in our experiments, whereas a gaseous mixture of 2.2% DADP in n-octane was used in work of Cafferata et al. The activation energies, however, appear to be very close: 159.2 kJ/mol (38.0 kcal/mol) in our work and 163.2 ± 10.5 kJ/mol (39.0 ± 2.5 kcal/mol) in Cafferata's one.

As is evident from Fig. 7, DADP is more stable peroxide than TATP. Its decomposition rate is approximately 2 times less than that of TATP. By and large, the studied peroxides are stable enough substances. Their thermal stability often exceeds stability of nitroesters and approaches to stability of alkyl azides [28].

3.3. Burning behavior and flame structure of TATP and DADP

Belyaev was the first who investigated the steady-state combustion of pressed TATP pellets in subatmospheric pressure region (0.008–0.1 MPa) [29]. The burning rate was shown to increase proportionally with pressure to reach approximately 1 cm/s at atmospheric pressure. Burning of TATP at high pressures was studied subsequently at Mendeleev University in the form of thin plates compacted to the maximum density [30]. The burning rate was measured to be 29 cm/s at 10 MPa that was comparable with level of burning of another primary explosive tetrazene.

Combustion of DADP pressed into 4 mm i.d. acrylic tubes was studied in the pressure interval of 0.05–2 MPa, in which the burning rate grows with pressure as r_b (mm/s) = 13.38·$P^{0.82}$. It follows from the comparison of burning rate-pressure dependences of DADP and TATP (Fig. 8) that diperoxide burns approximately 5 times slower that triperoxide.

Temperature profiles in the TATP and DADP combustion waves were obtained at pressures of 0.03, 0.05, 0.1, and 0.2 MPa. The temperature profiles for both peroxides reveal single-stage flame structure (Fig. 9), the maximum combustion temperature remained unchanged when pressure increased. An increase in pressure only led to some increase in the surface temperatures for both substances, although not clearly distinguishable on the temperature profiles. An important difference in the thermal behavior between DADP and TATP is that characteristic temperatures on profiles (T_s and T_f) for DADP are less than those for TATP (Table 5).

Since both DADP and TATP are known to be low-melting and volatile substances, their burning surface temperatures may be expected to correspond to the boiling points. In spite of a considerable scatter of the experimental data, the measured surface temperatures of TATP and DADP are generally located in the vicinity of the corresponding lines of their vapor pressure dependences (see Fig. 4).

3.4. Combustion mechanism of DADP and TATP

Because of the volatile nature of the studied peroxides, it has been suggested that their combustion obeys gas-phase combustion mechanism. Using a well-known combustion model for volatile

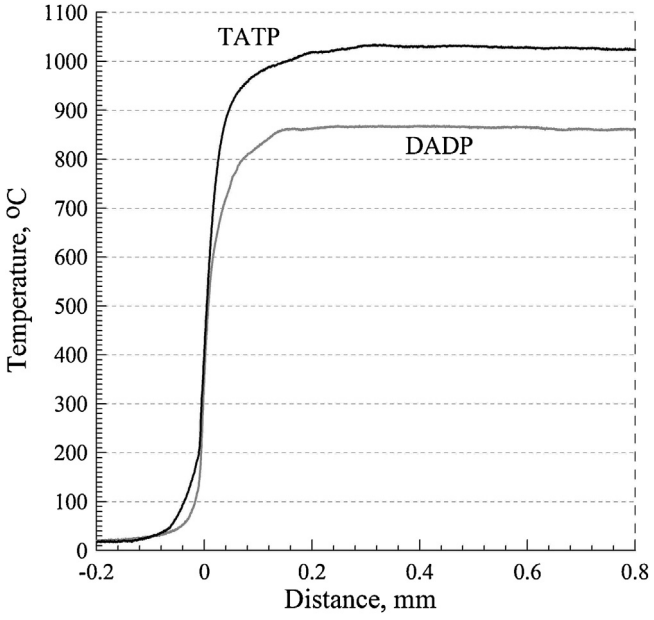


Fig. 9. Comparison of typical TATP and DADP temperature profiles at 0.1 MPa.

Table 5
Characteristic temperatures in the combustion wave of TATP and DADP.

Pressure (MPa)	TATP		DADP	
	Surface temperature, T_s (°C)	Maximum flame temperature, T_f (°C)	Surface temperature, T_s (°C)	Maximum flame temperature, T_f (°C)
0.03	166 ± 17	1014 ± 3	–	–
0.05	–	1020	130 ± 7	862 ± 3
0.1	179 ± 12	1029 ± 9	149 ± 12	869 ± 2
0.2	179 ± 12	1088 ± 35	162 ± 18	871 ± 3

explosives [31], one can easily calculate the formal kinetics of the gas-phase reaction which is responsible for heat generation in the gas and controls the burning rate. The combustion model employed links the kinetics of the dominant reaction under flame conditions with the rate of burning:

$$u_m = \sqrt{\frac{2\lambda\mu}{c_p(T_f - T_0)}\left(\frac{RT_f{}^2}{E}\right)^{n+1}\frac{n!}{(T_f - T_{01})^n}\left(\frac{P}{RT_f}\right)^n A_n e^{-E/RT_f}},$$

where u_m is the mass burning rate; T_f is the flame temperature; μ is the molecule weight of peroxide; p is pressure; E, A_n, and n are the activation energy, preexponential factor, and the formal order of the leading reaction; c_p (0.49 cal/g K) and λ (0.0002 cal/cm s K) are the average specific heat and thermal conductivity of the gas phase. The value of $T_{01} = T_0 - (L_m + L_{vap})/c_p$ stands for the initial temperature, T_0, with allowance for the heat consumed for melting, L_m, and vaporization, L_{vap}. Thermophysical properties were calculated with the use of the "REAL" computer code for simulation of chemical equilibriums [17].

It is clear from Fig. 10 that kinetic parameters of the reactions derived from the combustion data are in correspondence with kinetic parameters of low-temperature thermal decomposition if extrapolated to the high-temperature zone. Therefore, it is safe to say, that the leading reactions in combustion of both peroxides are their decomposition reactions occurring under flame conditions. This may be caused by relatively low flame temperatures of both peroxides, at which secondary radical reactions with low activation energy are still faster than the decomposition reactions with high activation energy [32,33].

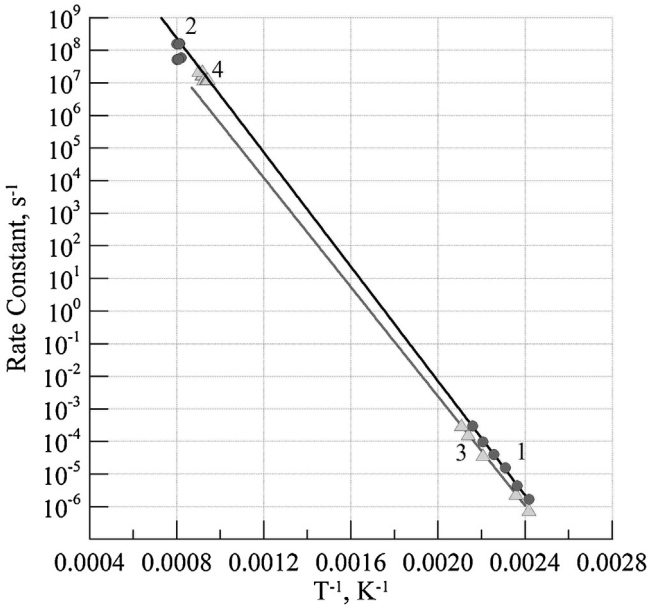


Fig. 10. Comparison of rate constants of the leading reaction in combustion (2, 4) and rate constants of gas-phase decomposition (1, 3) for TATP (1, 2) and DADP (3, 4).

3.5. Comparative estimation of impact sensitivity of DADP and TATP

Impact sensitivity of DADP and TATP was measured using K-44-1 impact machine for primary explosives in comparison with a known primary explosive tetrazene. The peroxide samples were prepared by pressing dry crystalline substances as received after crystallization at 100 MPa. Neither crystal shape nor size was specially checked. The Bruceton analysis method was used to determine H_{50} values. The measured H_{50} values for DADP, TATP, and tetrazene, taken for comparison purposes, are 21.2 ± 1.5 cm, 11.7 ± 0.8 cm, and 16 ± 2 cm, respectively. DADP appeared to be less sensitive to impact than TATP. The latter in its turn is more sensitive than tetrazine, one of the most sensitive primary explosive used for sensitization of many priming compositions. It should be noted that DADP is also less sensitive than TATP to friction as well as it was recently shown by Matyáš et al. [34].



Fig. 11. Initiating efficiency test results: 0.1 g of TATP (a) and 0.5 g of DADP (b) pressed at 30 MPa.

3.6. Initiating efficiency of DADP

Surprisingly, none of DADP samples of 0.1, 0.2, 0.3, and 0.5 g pressed at 30 MPa was able to induce detonation in the secondary RDX charge. Under the same test conditions a detonator charged with a sample of TATP as small as 0.1 g made a full-size hole in a led witness plate (Fig. 11).

Such different initiating behaviors of two allied substances are obviously due to different rates of burning at normal and elevated pressures first, and different burning temperatures, second. These parameters are known to be responsible for deflagration-to-detonation transition length [35]. This transition length is probably significantly more in DADP pressed charge than in TATP.

4. Conclusion

A comparative investigation of vapor pressure, thermal decomposition, burning behavior, impact sensitivity, and initiating efficiency of two potentially initiating explosive compounds, diacetone diperoxide and triacetone triperoxide, has been carried out.

The heats of DADP and TATP evaporation and the heat of DADP sublimation were obtained, and the heat of TATP sublimation was adjusted. The DADP vapor pressure was found to be significantly higher than that of TATP.

The study has shown that the decomposition rate constant of DADP is approximately two times less than that of TATP. On the whole, the thermal stability of the studied peroxides is higher than the thermal stability of nitroesters and approaches to that of alkyl azides.

The linear burning rate of DADP was measured to be approximately five times less than that of TATP. Thermocouple measurements showed that a single-stage flame is observed on combustion of both peroxides, with the maximum flame temperature somewhat lowered in the case of DADP.

The leading reaction on combustion of both peroxides was proposed to occur in the gas phase. The kinetic parameters of this reaction derived from combustion data corresponded to the kinetic parameters of low-temperature thermal decomposition if extrapolated to the high-temperature zone. As compared with TATP, DADP has less flame temperature and less de-composition rate, which are reasons for its lowered burning rates.

If comparing explosive properties of the studied peroxides, DADP may be considered as a less dangerous in handling explosive as TATP. It showed notably less sensitiveness to drop-weight impact than TATP, and no deflagration-to-detonation transition was observed when RDX was attempted to explode by 0.5 g of DADP in the standard detonator shell.

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